

EXPERIMENT -09

Problem Statement

Linux systems use a unified directory structure in which storage devices such as hard disks, USB drives, and external storage must be explicitly attached to the file system to access their contents. Unlike some operating systems, where devices are automatically available, Linux requires administrators to understand how devices are identified, mounted, and unmounted.

The problem is to identify storage devices in Linux, mount and unmount file systems manually, and understand the significance of mounting points and device management in system administration and cybersecurity environments.

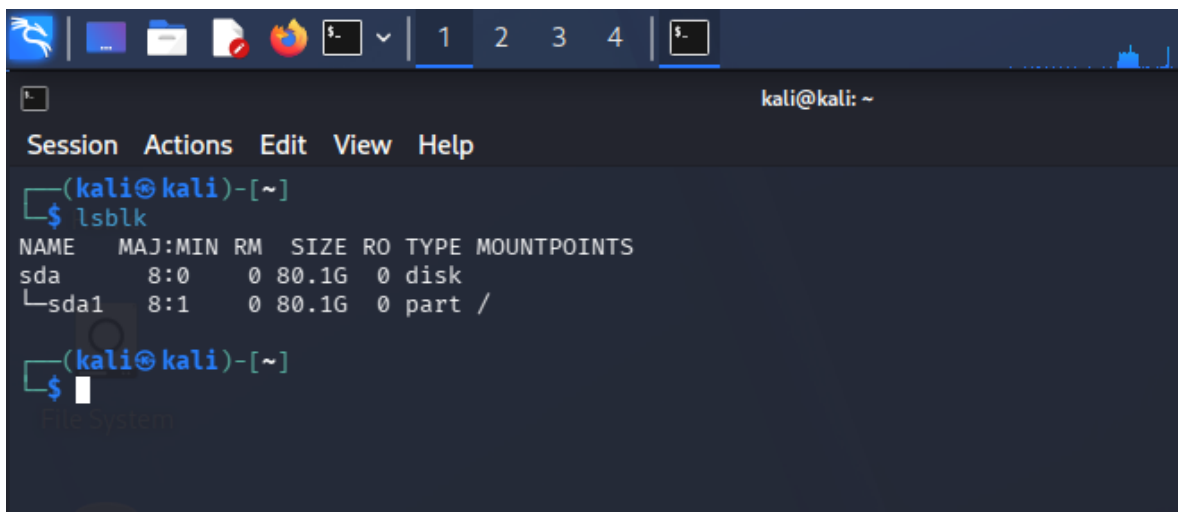
AIM - Mounting and Unmounting Devices in Linux

Procedure

Identifying Devices

Open the terminal.

List available block devices: `lsblk`



```
(kali㉿kali)-[~]
$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda           8:0    0 80.1G  0 disk
└─sda1        8:1    0 80.1G  0 part /
```

The screenshot shows a terminal window with the title bar "kali@kali: ~". The terminal has a menu bar with "Session", "Actions", "Edit", "View", and "Help". The user is in the home directory (~) and has executed the command `lsblk`. The output is a table of block devices. The first device is `sda`, which is 80.1G in size, has 0 read-only flags, and is of type `disk`. It has a single partition `sda1`, which is also 80.1G in size, has 0 read-only flags, and is of type `part`, mounted at the root `/`.

Display detailed disk information: `sudo fdisk -l`

```
(kali@kali)-[~]
$ sudo fdisk -l
[sudo] password for kali:
Disk /dev/sda: 80.09 GiB, 86000000000 bytes, 167968750 sectors
Disk model: VMware Virtual S
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x7ae167e4

Device      Boot Start      End  Sectors  Size Id Type
/dev/sda1   *      2048 167968749 167966702 80.1G 83 Linux

(kali@kali)-[~]
$
```

Creating a Mount Point

Create a directory to be used as a mount point: `sudo mkdir /mnt/usb`

Mounting a Device

Mount a partition (example `/dev/sdb1`) to the mount point: `sudo mount /dev/sdb1 /mnt/usb`

```
(kali@kali)-[~]
$ sudo mkdir /mnt/usb

(kali@kali)-[~]
$ sudo mount /dev/sdb1/mnt/usb
mount: /dev/sdb1/mnt/usb: can't find in /etc/fstab.

(kali@kali)-[~]
$
```

Verify the mount: `df -h mount`

```
(kali@kali)-[~]
$ df -h mount
df: mount: No such file or directory

(kali@kali)-[~]
$
```

Accessing the Mounted Device

Navigate to the mounted directory: `cd /mnt/usb`

```
(kali㉿kali)-[~]
$ cd /mnt/usb

(kali㉿kali)-[/mnt/usb]
$
```

Unmounting the Device

Exit the mounted directory: `cd ~`

```
(kali㉿kali)-[/mnt/usb]
$ cd ~

(kali㉿kali)-[~]
$
```

Unmount the device: `sudo umount /mnt/usb`

Verify unmounting: `df -h`

```
(kali㉿kali)-[~]
$ sudo umount /mnt/usb
sudo: umount /mnt/usb: command not found

(kali㉿kali)-[~]
$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            879M   0    879M   0% /dev
tmpfs           197M  1.3M  196M   1% /run
/dev/sda1       79G   16G   59G  22% /
tmpfs           982M  4.0K  982M   1% /dev/shm
none            1.0M   0    1.0M   0% /run/credentials/systemd-journald.service
tmpfs           982M  20K   982M   1% /tmp
none            1.0M   0    1.0M   0% /run/credentials/getty@tty1.service
tmpfs           197M  116K  197M   1% /run/user/1000
none            1.0M   0    1.0M   0% /run/credentials/getty@tty2.service

(kali㉿kali)-[~]
$
```

Expected Output

- Storage devices and partitions are listed successfully.
- Device is mounted to the specified mount point.
- Files inside the mounted device are accessible.
- Device is safely unmounted without errors.

Result

Storage devices were successfully identified, mounted, accessed, and unmounted in Linux. The student gained practical understanding of Linux file system mounting and device management.